# DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY LUCKNOW




## Evaluation Scheme & Syllabus

**for**

## M. Tech.

## ARTIFICIAL INTELLIGENCE & DATA SCIENCE

**(Effective from the Session: 2021-22)**

## M.Tech (Artificial Intelligence and Data Science)

### Objectives:

M.Tech in Artificial Intelligence and Data Science program provides students with a state-of-the-art educational experience to become outstanding, socially responsible engineers of AI systems. Students will learn the theoretical foundations and gain practical skills to manage and engineer various data forms, to make predictive analyses, to derive actionable insights, and develop intelligent hardware and software systems and related cutting-edge AI technologies. This program also provides students with foundational knowledge, experience, and skills that are crucial for a data science career. It features a multidisciplinary curriculum in various application domain of data science such as data Science in health care, social networks, public policy and finance through applied math, statistics and machine learning. Its rigorous, balanced coursework provides students with strong mathematical, computing, and data skills, as well as project experience.

Graduates of this program will have the necessary theoretical knowledge, practical data analysis and coding skills for solving complex real-world problems based on massive amounts of data and also meet the qualifications for various data scientist positions in industry and government. They will graduate with necessary training in applied research in the area.

### Program Learning Outcomes

1. Identify and apply fundamental concepts, algorithms, and methodologies to formulate, analyze, design, and evaluate artificial intelligence and machine learning systems;
2. Apply computer science knowledge and tools to assist in performing data science tasks. Summarize and evaluate statistical and machine learning concepts, models, and techniques
3. Tackle complex machine learning and artificial intelligence problems using contemporary principles, algorithms, technologies, methodologies, and tools;
4. Participate in a team to develop AI and machine learning applications; while recognizing the ethical, economic and environmental implications of their work;
5. Sustain a process of life-long learning in engineering or other professional areas; and Identify ways in which data scientists can contribute to the cultural and economic well-being of diverse societies in local, national, and global scopes.

**M.Tech. (Artificial Intelligence and Data Science)**

## Semester– I

| Code | Credit | Core/Elective | Subject |
|---|---|---|---|
| MTDS-101 | 3 | Core Subject | Analysis of Algorithms |
| MTDS-102 | 3 | Core Subject | Artificial Intelligence |
| Department Elective-I | 3 | Elective | Elective-A |
| Department Elective-II | 3 | Elective | Elective-B |
| MTCC-101 | 3 | Core Subject | Research Process and Methodology |
| MTDS-151 | 2 | Programming Laboratory | Programming Lab (Assignments based on Algorithm/AI) |
| MTDS-152 | 1 | Project based learning | Mini Project/ Term Paper |
| **Total Credit** | **18** | | |

## Semester– II

| Code | Credit | Core/Elective | Subject |
|---|---|---|---|
| MTDS-201 | 3 | Core Subject | Numerical and Statistical Methods for Data Science |
| MTDS-202 | 3 | Core Subject | Database Systems |
| Department Elective-III | 3 | Elective | Elective-C |
| Department Elective-IV | 3 | Elective | Elective-D |
| Department Elective-V | 3 | Elective | Elective-E |
| MTDS-251 | 2 | Programming Laboratory | Programming Lab (Assignments based on Data Science/ML /Database)) |
| MTDS-252 | 1 | Project based learning | Mini Project/ Term Paper |
| **Total Credit** | **18** | | |

## Semester– III

| Code | Credit | Core/Elective | Subject |
|---|---|---|---|
| MTDS-301 | 3 | Core Subject | Seminar |
| MTDS-302 | 15 | Core Subject | Dissertation/Internship |
| **Total Credit** | **18** | | |

## Semester– IV

| Code | Credit | Core/Elective | Subject |
|---|---|---|---|
| MTDS-401 | 20 | Dissertation/ Thesis | Dissertation/Internship |
| **Total Credit** | **18** | | |

## Elective Group– I (Machine learning/Natural Language Processing)

MTDS-011 MachineLearning
MTDS-012 Data Mining
MTDS-013 Natural Language Processing
MTDS-014 Pattern Recognition
MTDS-015 Big Data Engineering and Analytics
MTDS-016 DataAnalytics
MTDS-017 Virtual and Augmented Reality

## Elective Group-II (AIand Web learning)

MTDS-021 Deep Learning
MTDS-022 Intelligent AutonomousSystems
MTDS-023 AI Tools and Practice for Systems Engineering
MTDS-024 Web and Big Data Mining
MTDS-025 Optimization Theory and Techniques

## Elective Group-III ( Data Science for Public Policy/Business Analytics/Financial Engg)

MTDS-031 Data Science Theory and Practice
MTDS-032 Data Visualization
MTDS-033 Foundations of Policy Analysis
MTDS-034 Multivariate Statistical Analysis
MTDS-035 Data Science for Biomedical Systems
MTDS-036 Business Data Analytics
MTDS-037 DataSciencein Finance
MTDS-038 Game theory for AI and Data Science

**Elective Group–IV (Data Science for Social Networks)**

MTDS-041 Analysis of Large-Scale Social Networks
MTDS-042 Knowledge Representation and Reasoning
MTDS-043 Big Data Security
MTDS-044 Real-time AI Video Analytics
MTDS-045 Text Analytics
MTDS-046 Data Analytics for Social Media

**Elective Group–V(Data Science for Health Care)**

MTDS-051 Health Informatics
MTDS-052 Principles of BiologicalSystems
MTDS-053 Introductionto Biomedical Informatics
MTDS-054 Advanced Biomedical Imaging
MTDS-055 Integration of Medical ImagingSystems
MTDS-056 Medical Diagnostics, Therapeutics and Informatics Applications
MTDS-057 Health Service Delivery
MTDS-058 Principles of Epidemiology

**Elective Group–VI (Mathematics forAI and Data Science)**

MTDS-061 Advanced Numerical Methods
MTDS-062 Numerical Analysis and Computation
MTDS-063 Applied Probability
MTDS-064 Optimization Theory, Algorithms and Techniques
MTDS-065 Theory and Computational Methods for Optimization
MTDS-066 Applied Matrix Analysis

**Note**:
*Candidates are expected to take electives from above six (06) specialization elective groups as per his/her interest. Candidates are allowed to take upto two electives from a single group. All other electives must be from different groups*

## Semester– I: Evaluation Scheme

| Code | Subject | Period | | | Credit | Evaluation Scheme | | | | | Subject Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | L | T | P | | Theory | | | Practical | | |
| | | | | | | CT | TA | ESE | TA | ESE | |
| MTDS-101 | Analysis of Algorithms | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| MTDS-102 | Artificial Intelligence | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| DE1 | Elective-A | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| DE2 | Elective-B | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| MTCC-101 | ResearchProcess and Methodology | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| MTDS-151 | Programming Lab (Assignments based on Algorithm/AI) | 0 | 0 | 3 | 2 | | | | 20 | 30 | 50 |
| MTDS-152 | Mini Project/ Term Paper | 0 | 0 | 2 | 1 | | | | 20 | 30 | 50 |
| **TOTAL** | | | | | **18** | | | | | | **600** |

## Semester– II: Evaluation Scheme

| Code | Subject | Period | | | Credit | EvaluationScheme | | | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | L | T | P | | Theory | | | Practical | | |
| | | | | | | CT | TA | ESE | TA | ESE | |
| MTDS-201 | Numerical and Statistical Methods for Data Science | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| MTDS-202 | Database Systems | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| Elective-III | Elective-C | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| Elective-IV | Elective-D | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| Elective-V | Elective-E | 3 | 0 | 0 | 3 | 20 | 10 | 70 | | | 100 |
| MTDS-251 | Programming Lab (Assignments based on Data Science/Database ) | 0 | 0 | 3 | 2 | | | | 20 | 30 | 50 |
| MTDS-252 | Mini Project/ Term Paper | 0 | 0 | 2 | 1 | | | | 20 | 30 | 50 |
| **TOTAL** | | | | | **18** | | | | | | **600** |

## Semester–III: Evaluation Scheme

| Code | Subject | Period | | | Credit | Evaluation Scheme | | | | | Subject Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Theory | | | Practical | | |
| | | L | T | P | | CT | TA | ESE | TA | ESE | |
| MTDS-301 | Seminar | | | 6 | 3 | | | | 100 | | 100 |
| MTDS-302 | Dissertation | | | 30 | 15 | | | | 200 | 300 | 500 |
| **TOTAL** | | | | | **18** | | | | | | **600** |

## Semester–IV: Evaluation Scheme

| Code | Subject | Period | | | Credit | Evaluation Scheme | | | | | Subject Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Theory | | | Practical | | |
| | | L | T | P | | CT | TA | ESE | TA | ESE | |
| MTDS-401 | Dissertation | | | 36 | 18 | | | | 200 | 400 | 600 |
| **TOTAL** | | | | | **18** | | | | | | **600** |

# SYLLABUS

| MTDS101 ANALYSIS OF ALGORITHMS | | |
|---|---|---|
| **Course Outcome(CO)** | | **Bloom's Knowledge Level (KL)** |
| **At the end of course, the student will be able to understand** | | |
| CO1 | Understand elementary concepts of Algorithms | K1, K2 |
| CO2 | Understand and apply Divide and conquer algorithms, greedy methods, dynamic programming and backtracking approach for problem solving | K2,K3 |
| CO3 | Discuss and Understand Genetic Algorithms and Artificial Neural Networks | K1, K2 |
| CO4 | Discuss , Understand and analyse complexity theory | K1, K2,K4 |
| CO5 | Discuss and Understand Randomized and Approximation Algorithms and solve real world problems | K1, K2, K5 |

DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| I | **Fundamentals:** Introduction to Algorithms, Iterative Algorithm, Design Issues**,** Computation Models and Design by Refinements Proof Rules-Basics Design by Proof Rules ,Design using recursion. | **08** |
| II | **Abstract Algorithms:** Divide and Conquer, Greedy methods Dynamic Programming, Backtracking, Branch and Bound method, Elementary Graph Algorithms | **08** |
| III | **Natural Algorithms:** Genetic Algorithms, Simulated Annealing, Artificial Neural Networks | **08** |
| IV | **Complexity Theory:** Introduction to Complexity theory ,Diagonalization Theorem ,Tractable and Non-Tractable problems; few NP and NPC problems | **08** |
| V | **Advanced Topics:** Randomized and Approximation Algorithms ,Introduction to Computational Geometry | **08** |

## Textbooks & References:


1. Introduction to Algorithms by Cormen, Licerson, Steina and Rivest
2. Design and Analysis of Algorithm, by Aho, Hopcroft and Ullman
3. Algorithm Design by Kleinberg and Tardos
4. Algorithms by Dasgupta, Papadimitriou and Vazirani
5. Introduction to Linear Optimization by Dimitris Bertsimas
6. The Design of Approximation Algorithms by Will iamson and Shmoys
7. Approximation Algorithms by Vazirani
8. Randomized Algorithms by Motwania and Raghavan
9. Scientific Articles published in International Journals and Conferences related to Data Analytics

**MTDS102 ARTIFICAL INTELLIGENCE**

| CourseOutcome(CO) | | Bloom's Knowledge Level (KL) |
|---|---|---|
| **At the end of course, the student will be able to understand** | | |
| CO1 | Understand the concepts of Artificial Intelligence and intelligent agents. | K1, K2 |
| CO2 | Understand and learn knowledge representation and reasoning for the problem-solving solving. | K1, K2,K3 |
| CO3 | Apply basic search techniques for problem-solving. | K3 |
| CO4 | Understand and apply learning techniques. | K1, K2,K3 |
| CO5 | Apply and utilize AI knowledge for application in the real world. | K3, K5, K6 |

## DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| **I** | Introduction to Artificial Intelligence, Foundation and history of Artificial Intelligence, Agents and Environments, Structure of intelligent agents, State-of-the-art applications of AI.<br><br>Agents and Environments, Structure of intelligent agents, State-of-the-art applications of AI. | **08** |
| **II** | Knowledge Representation and Reasoning: Representation and Reasoning using predicate logic, Inference in first order logic, forward and backward chaining. Probabilistic reasoning, Bayesian networks, Dempster Shafer theory, Probabilistic Reasoning over time: Hidden Markov Models, Kalman Filters. | **08** |
| **III** | Uninformed search and Informed search based on heuristics, Local search algorithms and optimization problems, Adversarial search: Games, Optimal decisions in games, Alpha-beta pruning, Online search. | **08** |
| **IV** | Learning from examples, Forms of Learning, Inductive Learning, Learning decision trees, Learning in problem solving, Learning Probabilistic models, Bayesian learning, Learning in neural and belief networks. Learning with hidden variable. | **08** |
| **V** | AI Applications: Expert system, decision support systems, speech and vision, natural language processing, semantic web, robotics, Al-based programming Tools. | **08** |

## Textbooks & References:


1. Artificial Intelligence: A Modern Approach, S Russel and P Norvig, 3rd Edition, 2015 Prentice Hall.

2. Introduction to Artificial Intelligence and Expert Systems, Dan W. Patterson, Pearson Education.

3. Artificial Intelligence and Expert Systems — Patterson PHI.

4. Research literature

**MTDS011 MACHINE LEARNING**

| Course Outcome (CO) | | Bloom's Knowledge Level (KL) |
|---|---|---|
| **At the end of course, the student will be able to understand** | | |
| CO1 | Discuss various concepts of machine learning with hypothesis and version space | K1, K2 |
| CO2 | Understand a wide variety of learning algorithms and how to evaluate models generated from data | K1,K3,K5 |
| CO3 | Understand and apply Bayesian method of machine learning | K1, K2,K3 |
| CO4 | Understand and apply Instance based learning and Case based learning | K1,K2,K3 |
| CO5 | Apply and evaluate machine learning algorithms on real world | K2, K3,K5 |

## DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| I | Introduction: Well defined learning problems, Designing a Learning System, Issues in Machine Learning, THE CONCEPT LEARNING TASK-General-to-specific ordering of hypotheses, Find-S, List then eliminate algorithm,Candidate elimination algorithm,Inductive bias. | 08 |
| II | Decision tree learning algorithm-Inductive Bias-Issues Decision tree learning; ARTIFICIAL NEURAL NETWORKS – Perceptrons, Gradient descent and the Delta Rule, Adaline, Multilayer Networks, Derivation Of Backpropagation Relu Backpropagation Algorithm Convergence, Generalization. | 08 |
| III | Estimating Hypotheses Accuracy, Basics of sampling Theory, Comparing Learning Algorithms; Bayesian Learning: Bayes Theorem, Concept Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, EMalgorithm. | 08 |
| IV | Sample Complexity for Finite Hypothesis spaces, Sample Complexity for Infinite Hypothesis spaces, The Mistake Bound Model of Learning; INSTANCE-BASED LEARNING–k- Nearest Neighbour Learning, Locally Weight dR egression, Radial basis function networks, Case-based learning. | 08 |
| V | Anillustrative example, Hypothesis spaces search, Genetic Programming, Models of Evolution and Learning; Learning first order rules-sequential covering algorithms General to specific be amsearch-FOIL; REINFORCEMENT LEARNING-The Learning Task, Q Learning. | 08 |

## Textbooks & References:


1. Tom M. Mitchell,— Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, Introduction to Machine Learning (Adaptive Computation and Machine Learning,The MIT Press 2004.
3. Stephen Marsland,—Machine Learning: An Algorithmic Perspective, CRC Press,2009.
4. Bishop,C.,Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.
5. Scientific Articles published in International Journals and Conferences related to Data Analytics

# MTDS016 DATA ANALYTICS

| Course Outcome (CO) | | Bloom's Knowledge Level (KL) |
|---|---|---|
| **Attheendof course, the student will be abletounderstand** | | |
| CO1 | Discuss various concepts of data analytics pipeline | K1, K2 |
| CO2 | Discuss and apply various data analytics methods | K1,K2,K3 |
| CO3 | Discuss and apply text and sentiment analysis | K1, K2,K3 |
| CO4 | Apply R tool for Data Analytics problem solving | K3,K5,K6 |
| CO5 | Understand NoSQL and Data Visualisation methods and implementation in tool | K1, K2,K5 |

## DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| I | Introduction: Data, Types of Data, Big Data, Big data Characteristics, Business Intelligence, Levels of measurement, Introduction to Statistical Learning, Mean, Median, Mode, Standard deviation. Life cycle of Data centric projects. | 08 |
| II | Basic Analysis Techniques: Chi-Square, t-Test, Correlation Analysis, Analysis of Variance. Advanced Analytics Techniques: Regression, Clustering, Classification, Association Mining. | 08 |
| III | Text Analytics & Web Mining: Process of Text Analytics, Topic Modelling, Sentiment Analysis, Web Mining. Time Series Analysis: Overview of Time Series Analysis, Forecasting Models, ARMA and ARIMA Models | 08 |
| IV | R language: Introduction to R., Basic Syntax, Implementation basic and advanced Data analytic methods, Data visualization using R, Text Analysis Process in R. | 08 |
| V | No SQL: Introduction to No SQL, Principles of No SQL Data Models, CAP, No SQL Data Model. Data Visualization: Plots and graphs for data visualization | 08 |

## Textbooks & References:


1. Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer
2. An and Rajaraman and Jeffrey David Ullman, Mining of Massive Datasets, Cambridge University Press.
3. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley
4. David Dietrich, Barry Heller, Beibei Yang, "Data Science and Big Data Analytics", EMC Education Series, John Wiley
5. Mark Gardner, "Beginning R: The Statistical Programming Language", Wrox Publication
6. Glenn J. Myatt, Making Sense of Data, John Wiley& Sons
7. Jiawei Han, Miche line Kamber "Data Mining Concepts and Techniques", Second Edition, Elsevier
8. Scientific Articles published in International Journals and Conferences related to Data Analytics

| MTDS-201 Numerical and Statistical Methods for Data Science | | |
|---|---|---|
| **CourseOutcome(CO)** | | **Bloom's Knowledge Level (KL)** |
| **At the end of course,the student will be able to:** | | |
| CO1 | Will be able to understand and apply the numerical techniques of interpolation, spline interpolation, linear and nonlinear regression to the problems. | K2,K3,K4 |
| CO2 | Will be able to solve the system of equation using numerical methods, develop understanding of ill conditioned equations and vectors space | K2,K3, K4 |
| CO3 | Will have understanding of probability theory, random processes and discrete and continuous probability distributions and be able to apply to the problems | K2, K3 |
| CO4 | Will be able to solve problems related to testing of hypotheses | K2,K3 |
| CO5 | Will develop understanding of group theory, important theorems and be able to apply Fermat's theorem and Chinese remainder theorem in solving the problems | K2, K3,K4 |

## DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| **I** | Interpolation: Polynomial and Cubic Spline Interpolation, Error analysis, Principle of least squares Regression: Polynomial, Logistic and Nonlinear regression, Gram Schmidt Process Orthogonal Transformation,MultipleRegression | **08** |
| **II** | System of Linear algebraic equation: Gaussian Elimination, Cholesky Decomposition, Ill-conditioned equations, Iterative Methods, Convergence, EigenValue problems, QR decomposition Vector space, subspaces, spanning set, linear transformation, Linear dependence and independence, basis and dimension | **08** |
| **III** | Probability, Statistics and Random Processes: Frequency Distribution Measures of central tendency, Probability theory, axioms, conditional probability, Bayes Theorem, Bernoulli Trials, Random variables, Probability distributions with discrete and continuous random variables:, geometric, Poisson, Normal, Exponential and Gaussian distributions,Expectations and Moments | **08** |
| **IV** | Sampling & Estimation Theory**:** Sampling and Testing of Hypothesis: Introduction To testing of hypothesis - Tests of significance for large samples – t, F and Chi-square tests; Estimation Theory, goodness of fit,Confidence interval, Correlation Coefficient | **08** |
| **V** | NumberTheory:Euclidean Algorithm,Cyclic Group Subgroup,Generators,Lagrange Theorem, Euler's Totient Function, Fermat's theorem, Chinese Remainder Theorem,Discrete Logarithm Problem | **08** |

**Textbooks References:**


**1.**Gilbert Stang: Introduction to Linear Algebra
**2.**Jain M.K.,Iyengar S.R.K.& Jain R.K.,Jain and Iyengar: Numerical Methods
**3**.Johnson R.A.: Probability and Statistics for Engineers
**4**.Papoulis A., Unnikrishna Pillai S.: Probability, random variable and stochastic Processes

| MTDS-202 Data Base Systems | | |
|---|---|---|
| **CourseOutcome (CO)** | | **Bloom'sKnowledgeLevel(KL)** |
| **At the end of course, the student will be able to understand** | | |
| CO1 | Understandand implement the concepts of database systems in real world,relational model and ER diagram. | K1, K2, K6 |
| CO2 | Apply query processing techniques to automate the real time problems of databases and understand and implement normalization | K2, K3,K4 |
| CO3 | Understand the concepts of transactions and database management issues including data integrity, security and recovery. | K1, K2,K3 |
| CO4 | Understand the concepts of distributed systems and techniques for message ordering | K1,K2,K3 |
| CO5 | Understand the concepts of distributed transactions and concurrency control in distributed environment | K1, K2,K3 |

| DETAILED SYLLABUS | | |
|---|---|---|
| **Unit** | **Topic** | **Proposed Lecture** |
| **I** | Data Modeling using the Entity Relationship Model, diagram, Relational data model concepts, integrity constraints, relational algebra, ERD, DFD using tools, e.g. Power designer | **08** |
| **II** | Database Systems Design and SQL: Querying Database Using SQL, Database Design Concepts, FDs, MVDs, JD and concepts in Normalization. | **08** |
| **III** | TransactionandConcurrencyControl,Schedules,Testingserializabilityofschedules,conflict&viewserializable,recoverability, Recovery from transaction failures, log basedrecovery,checkpoints,deadlockhandling. | **08** |
| **IV** | Theoretical Foundation for Distributed System, Logical Clocks ,Lamport's Vectorslogical clocks.<br>Message Passing Systems: causal order, totalorder, total causal order,Techniques for Message Ordering | **08** |
| **V** | Distributed Transactions, Commit protocols, Concurrency control in distributed transactions, Distributed Deadlocks, Transaction Recovery.Replication:System Modeland Group Communication,Fault-tolerant services | **08** |

**Textbooks & References:**


1. Korth, Silbertz, Sudarshan,"Database Concepts", McGraw Hill
2. Date,"An Introduction to Database Systems",Addison Wesley
3. Elmasri, Navathe,"Fundamentals of Database Systems", Addison Wesley
4. Singhal and Shivaratri, "Advanced Conceptin Operating Systems", McGraw Hill
5. Ramakrishnan, Gehrke," Database Management Systems", McGraw Hill
6. Coulouris, Dollimore, Kindberg,"Distributed System:Concepts and Design", Pearson Education


## MTDS-021 DeepLearning

| Course Outcome (CO) | | Bloom's Knowledge Level (KL) |
|---|---|---|
| **At the end of course,the student will be able to understand** | | |
| CO1 | Understand and Analyse concepts of Bayesian Learning, Neural networks, Perceptron convergence theorem , Backpropagation | K2, K4 |
| CO2 | Understand and Apply Deep Learning, CNN and Transfer Learning | K2,K3 |
| CO3 | Understand and analyse PCA And Deep learning architectures. | K2, K4 |
| CO4 | Explain and Understand Gradient Descent, Normalization, LSTM | K2 |
| CO5 | Implementation of Case Studies in Deep Learning | K3,K6 |

### DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| I | Introduction to Bayesian Learning, Decision Surface, Linear Models (SVM, Perceptron, Logistic Regression), Optimization Techniques, Gradient Descent, Linear and non linear separable problems, Perceptron Convergence Theorem, Introduction to Neural Network. | 08 |
| II | History of Deep Learning, Introduction to Probability, Continuous and Discrete Distributions, Probabilistic Theory of Deep Learning, Deep Vs Shallow Networks, Multilayer Perceptron, Back Propagation Learning, Convolutional Neural Network (CNN), Building Blocks CNN, Transfer Learning | 08 |
| III | Principal Component Analysis (PCA), Autoencoders, Deep Learning Architectures-AlexNet, VGG, ResNet, etc. | 08 |
| IV | Revisiting Gradient Descent, Momentum Optimizer, RMSPrOP, ADAM, Loss Functions, Effective Training Deep Net-Early Stopping, Drop-Out, Batch Normalization, Instance Normalization, Group Normalization, Recurrent Networks, LSTM. | 08 |
| V | ImageNet,WaveNet, Generative Modelling with DL, Generative Adversarial Networks, Case studies of DL models with some real life applications. | 08 |

**Textbooks & References:**


1. Ian Good fellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016.
2. Michael Nielsen, Neural Networks And Deep Learning, Determination Press, 2015.
3. Pattern Classification-Richard O.Duda, Peter E. Hart, David G.Stork, John Wiley & SonsInc.
4. Cosma Rohilla Shalizi, Advanced Data Analysis from Elementary Point of View, 2015
5. Deng Yu, Deep Learning: Methods and Applications, Now Publishers, 2013


## MTDS-024: WEB and BIG DATA MINING

| CourseOutcome(CO) | | Bloom'sKnowledgeLevel(KL) |
|---|---|---|
| **Attend Course,the student will be able to understand** | | |
| CO1 | Discuss and explain various concepts of Data Mining and Web Mining | K1, K2,K3 |
| CO2 | Discuss and explain various methods of Supervised and Unsupervised learning methods | K1,K2,K3 |
| CO3 | Discuss and explainInformation Retrieval and Web Search | K1, K2,K3 |
| CO4 | Understand Web crawling and implementation of Web Crawler and Page Rank Algorithm | K3,K5,K6 |
| CO5 | Discuss and explain Sentiment Analysis and Opinion Mining | K1, K2 |

### DETAILED SYLLABUS

| Unit | Topic | Proposed Lecture |
|---|---|---|
| I | Introduction to Web Data Mining and Data Mining Foundations, Introduction – World Wide Web (WWW), A Brief History of the Web and the Internet, Web Data Mining - Data Mining,Web Mining.Data Mining Foundations – Association Rules And Sequential Patterns – Basic Concepts of Association Rules, Apriori Algorithm - Frequent Item set Generation, Association Rule Generation, Data Formats for Association Rule Mining, Mining with multiple minimum supports, Mining Sequential Patterns on GSP, Mining Sequential Patterns on Prefix Span, GeneratingRulesfromSequentialPatterns. | 08 |
| II | Supervised and Unsupervised Learning Supervised Learning – Basic Concepts, Decision Tree Induction – Learning Algorithm, Impurity Function, Handling of Continuous Attributes, Classifier Evaluation, Rule Induction – Sequential Covering, Rule Learning, Classification Based on Associations, Naïve Bayesian Classification, Naïve Bayesian Text Classification – Probabilistic Framework, Naïve Bayesian Model Unsupervised Learning – Basic Concepts, K-means Clustering–K-means Algorithm, Representation of Clusters, Hierarchical Clustering – Single link method, Complete link Method, Average Link Method,Strength and Weakness. | 08 |
| III | Information Retrieval and Web Search: Basic Concepts of Information Retrieval. Information Retrieval Methods – Boolean Model, Vector Space Model and Statistical Language Model, Relevance Feedback, Evaluation Measures,Text and Web Page Preprocessing – Stopword Removal, Stemming, Inverted Index and Its Compression – Inverted Index, Search using Inverted Index, Index Construction, Index Compression, Latent Semantic Indexing–Singular Value Decomposition, Query and Retrieval, Web Search, Meta Search, Web Spamming. | 08 |

| | | |
|---|---|---|
| **IV** | Analysis, Co-Citation and Bibliographic Coupling, PageRank Algorithm, HITS Algorithm, Community Discovery-Problem Definition, Bipartite Core Communities, Maximum Flow Communities, Email Communities. Web Crawling – A Basic Crawler Algorithm - Breadth First Crawlers, Preferential Crawlers, Implementation Issues – Fetching, Parsing, Stopword Removal, Link Extraction, Spider Traps, Page Repository, Universal Crawlers, Focused Crawlers, Topical Crawlers, Crawler Ethicsand Conflicts | **08** |
| **V** | Sentiment Analysis and Opinion Mining – Sentiment Classification – Classification based on Sentiment Phrases, Classification Using Text Classification Methods, Feature based Opinion Mining and Summarization –Opinion Search and Opinion Spam. Aspect Based Sentiment Analysis, Supervised and Unsupervised Spam Detection, Group Spam Detection. | **08** |

**Textbooks & References:**


1.Web Data Mining: Exploring Hyperlinks, Contents and usage data by Bing Liu (Springer Publication)

2.Data Mining: Concepts and Techniques, Second Edition by Jiawei Han, Micheline Kamber (Elsevier Publications)

3.Mining the Web: Discovering Knowledge from Hypertext Data by Soumen Chakrabarti (Morgan Kaufmann Publishers)

4.Research Articles & Journal Papers


## MTDS-064: Optimization Theory, Algorithms and Techniques

| **Course Outcome (CO)** | | **Bloom's Knowledge Level (KL)** |
|---|---|---|
| **At the end of course, the student will be able to understand** | | |
| CO1 | Discuss and explain various concepts of optimization methods. | K1, K2 |
| CO2 | Explain Linear programming and analyse problems. | K1, K2,K3 |
| CO3 | Discuss and explain Transportation and transhipment problem and Assignment Problem | K1, K2 |
| CO4 | Discuss , Explain and analyse InventoryModels and Dynamic programming problems | K1, K2,K3 |
| CO5 | Discuss ,explain and apply Queuing theory and Integer Programming | K2,K3,K5 |

### DETAILED SYLLABUS

| **Unit** | **Topic** | **Proposed Lecture** |
|---|---|---|
| **I** | Introduction to optimization techniques: Nature, scope, applications, Historical development, Role In decision making. Introduction to linear programming: Graphical approach with algorithm, Simple Method with algorithm, Duality, Post optimal analysis. | **08** |
| **II** | Linear Programming:Transportation Problems With Algorithms-North West Method, Column minima method, Row Minima method, Matrix minima method,Vogel's Approximation method, Modified method,Transhipment Problem. | **08** |
| **III** | Assignment Problems: Hungarian algorithm and its application. Dynamic Programming: Bellman's principle of optimality of Dynamic Programming, Application. | **08** |
| **IV** | Inventory Models: Various Models, Applications, Economic Lot Sizes - price. Replacement Problems: Concept, Applications. | **08** |

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|---|---|---|
| **V** | Queuing Theory: Different models and their solution. Integer Programming: Concept, Applications, cutting plane method,Branch And Bound Method | **08** |

**Textbooks & References:**


1. Hadley,G. ,"Linear Programming, and Massachusetts",Addison-Wesley
2. Taha, H.A,"Operations Research–An Introduction", Macmillian
3. Hillier, F.S.,G.J .Lieberman,"Introduction to Operations Research", Holden-Day
4. Harvey M.Wagner,"Principles of Operations search with Applications to Managerial
5. Decisions", Prentice Hall of India Pvt. Ltd.
6. Swaroop Ketal,"Operation Research", S.Chand


**Subject Code: MTDS-151** **Subject Name: Programming Laboratory**

| S.No. | Experiment / Assignment |
|---|---|
| 1. | Perform the algorithmic study of various classifierssuchas Bayesian classifiers, Decision Tree, Support Vector Machines. The study focuses on the design, implementation and performance comparison of different classifiers. |
| 2. | Perform the algorithmic study Markovian process.Designa Speech recognition system using Hidden Markov Model. |
| 3. | Perform an algorithmic study of convolution networks.Evaluated The performance ofFense Net and ResNet deep learning models using suitable datasets. |
| 4. | Implement dependent task scheduling via Knapsack / TSP / Genetic Algorithm. Discuss the efficiency of the implementation. |
| 5. | WriteaprogramtoimplementMaxflowMinCutproblemusingtheFord Fulkersonalgorithm.The implementation must exhibit limitations ofthe problem.Also should explain whether the algorithm works for bipartite matching or not. |
| 6. | Write a program to implement string matching using Rabin Karp algorithm, Knuth Morris algorithm, and Boyer Moore algorithm. |
| 7. | WriteaprogramtofindallpairshortestpathusingJohnson'salgorithmfor sparse graphs. |
| 8. | Chat bots are computer programs that simulate human conversation through text or voice interactions. Write a program to implement AI based chatbots. |

**Subject Code: MTDS-251** **Subject Name: Programming Laboratory**

| S.No. | Experiment/Assignment |
|---|---|
| 1. | Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample. |
| 2. | Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets. |
| 3. | Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets. |
| 4. | Considering "US Crime Dataset in "MASS " package in R, use t-test to analyze the difference between southern group (So=1) Andno southern group (So=0) on the probability of imprisonment (Probe). Discuss hypothesis and results. |
| 5. | Considering "Loblolly" dataset in R determine mean, median, quartiles of age and height. Determine the correlation between age and height. Also apply the chi-square test to determine whether there is an association between age and height of a tree. hypothesis and result. |
| 6. | Considering the "cholesterol" dataset in the "multicomp" package in R, use ANOVA to analyze the difference between trt and response. Discuss the hypothesis and result |
| 7. | Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set. |
| 8. | Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. |
| 9. | Write a program to implement k-Nearest Neighbour algorithm to classify the iris data set. Print both correct and wrong predictions. |
| 10. | Implement a recommender system-A recommender system is a tool that consider several metrics such as age, previously watched shows, most-watched genre, watch frequency, and feeds them into a Machine Learning model which then generates what the user might like to watch next. Various media platforms like YouTube, Netflix implement the system to recommend you what to watch next<br><br>The objective is to build a recommendation engine (like the ones used by Amazon and Netflix) that can generate personalized recommendations for products, movies, music, etc., based on customer preferences, needs, and online behaviour. |
| 11. | Implement a Web Crawler with the following functionality-<br><br>● The crawler collects a set of web pages and identify their properties related to the web structure<br>● Identifies a portion of the Web (a subtree, a server or a topic oriented part of the Web) to be analyzed.<br>● Ranking of pages by using various techniques.<br>● Grouping of pages by similarity. |

1. **The instructor can add/modify/tune experiments, wherever he/she feels in a justified manner.**
2. **The language/tools/environment to conduct the lab are R, Python, MATLAB/JAVA**
3. **All students are expected to complete the study and coding by the expected date of completion.**